



RS004 Week 1 — English Worksheet

Topic: Game Window & Game Loop · **Course:** Platformer Game · **Time:** about 45 minutes

This week you think about your very first **game window** and the **game loop** — the heartbeat that runs every frame: read events → update → draw. On paper you will read this code, predict it, fix it, and explain it in your own words. Everything you build for the next 16 weeks lives inside this loop.

Keep these words handy: **game loop**, **event**, `pygame.QUIT`, `flip()`, **FPS**, **RGB**.

1 · Predict 🧠

Read each snippet. Write what you think it will do.

```
screen.fill((135, 206, 235))
pygame.display.flip()
```

What colour fills the screen, and what does `flip()` do?


```
for event in pygame.event.get():
    if event.type == pygame.QUIT:
        running = False
```

A player clicks the X button. What happens to `running`, and then to the loop?


```
clock.tick(60)
```

This line is at the end of the loop. What is it controlling?

2 · Read & Match

Match each line to its job. Write the letter next to each number.

1. `pygame.init()`
2. `pygame.display.set_mode((800, 600))`
3. `pygame.display.set_caption("나의 플랫폼머")`
4. `screen.fill(SKY_BLUE)`
5. `pygame.display.flip()`

- A. paints the whole screen one colour
- B. starts up pygame so we can use it
- C. shows everything we drew this frame
- D. makes the 800×600 window
- E. writes the title on the window bar

1 → ___ 2 → ___ 3 → ___ 4 → ___ 5 → ___

3 · Read the Colours 🎨

Each colour is (Red, Green, Blue), where each number is 0–255. Write down what colour you think each one makes.

(255, 0, 0) → _____
(0, 0, 0) → _____
(255, 255, 255) → _____
(135, 206, 235) → _____

Now invent one. Write an RGB value for a sunset orange sky, and say what each of the three numbers means.

4 · Spot the Bug 🐛

Each snippet is broken. Read what it should do, then rewrite it so it works. Explain why the original was wrong.

Bug A — The window should stay open until the player closes it. Right now it flashes open and shuts instantly.

```
running = True
for event in pygame.event.get():
    if event.type == pygame.QUIT:
        running = False
screen.fill((135, 206, 235))
pygame.display.flip()
```

Hint: the drawing and event-reading need to *repeat*. What kind of loop is missing?

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The screen should turn sky-blue, but the window stays black even though `fill` runs.

```
screen.fill((135, 206, 235))
clock.tick(60)
```

Hint: we filled the screen but never *showed* it.

Write the fixed code:

Why was it wrong? Why does your fix work?

5 · Explain the Code

Here is a complete working game window — an 800×600 sky-blue screen running at 60 FPS. Read it carefully, then answer the questions.

```
import pygame

pygame.init()
screen = pygame.display.set_mode((800, 600))
pygame.display.set_caption("나의 플랫폼머")
clock = pygame.time.Clock()
SKY_BLUE = (135, 206, 235)

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
    screen.fill(SKY_BLUE)
    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

Which line creates the window, and how big is it?

What does the `while running:` loop do, and when does it stop?

Inside the loop, what is the job of `screen.fill(SKY_BLUE)` , and what is the job of `pygame.display.flip()` ?

What would change on screen if `SKY_BLUE` was `(255, 0, 0)` instead?

Why is `clock.tick(60)` inside the loop, and what does the 60 mean?

6 · Explain Your Lesson Code 🎥

Today in the live lesson you wrote your own game window. Take a short video on your phone and explain **the code you wrote** — you can show it running. Teach it like you are talking to someone who has never seen Pygame. Try to use these words: **game loop, event, fill, flip, FPS**.

1. Show your window opening.
2. Point at the line that sets the background colour and read it out.
3. Explain what your game loop does every frame.
4. Click the X and show the window closing — say which line made that work.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.